

30-10-2025 (Day-3)

1) Comments -

Comments are part of the code that are not executed by python.

They are mainly used to make code more readable and to give explanations about what the code does.

Types of comments :-

1) Single-line comment - starts with a #

This is a single line comment

```
print("Hello")
```

2) Multi-line comment - written inside triple quotes ''' or '''

```
'''This is a multi-line comment
```

```
Used when you want to describe logic or code in detail.'''
```

Note - Comments are ignored by the interpreter - they're for humans, not for machines.

2) Keywords -

Keywords are reserved words in Python that have a predefined meanings.

You cannot use them as variable names, function names, or identifiers.

ex- if, else, for, while, def, True, False, class, return, import, break, etc...

(Valid use) -

```
if True:
```

```
    print("This is a valid keyword use")
```

(Invalid use)

if = 10 # * Error! "if" is a keyword

3. Input & output functions -

output function $\rightarrow$ print()

$\hookrightarrow$ used to display data on the screen.

ex- `Print("Start Content Creation")`

Input function $\rightarrow$ input()

Used to take user input from the keyboard.

ex -

`name = input("Enter your name:")`

`Place = input("Enter your place:")`

`print("My name is", name, "and I am from", place)`

4. String Formatting Methods -

a) Concatenation method

`print("My name is" + name + ", I am from" + place)`

b) `format()` method

`print("My name is {}, I am from {}".format(name, place))`

c) f-string method

`print(f"My name is {name}, I am from {place}")`

$\hookrightarrow$ It's cleaner & faster compared to other methods.

5. Variables - A variable is a name that stores a value in memory. It acts like a container that holds data.

ex:-

`num = 15`

`name = "Poojtha"`

Valid variable name -

`A = 6`

`b = 12`

`num1 = 123`

`stu_name = "Poojtha"`

- sub-marks = 67.

(Invalid use)

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- sub-marks = 67.

Invalid variable names -

1num = 78 # cannot start with a number

stu id = 1234 # no spaces allowed

@name\$ = "vasu" # special symbols not allowed except - underscore

Assignment

1) What is Python and why is it called an interpreted language?

Python is high-level, object-oriented programming language that is used for developing websites, data analysis, machine learning, AI, automation, and more.

It is called an interpreted language because python code runs line by line using an interpreter instead of compiling the whole program at once.

If there's an error in one line, Python stops there and shows the error - this makes debugging easier.

2) What are the key features of python that make it popular for beginners & professionals?

→ Easy to learn - The syntax is simple and looks like English.

→ Interpreted language - No need for compilation; executes directly.

→ Platform Independent - Works on windows, Mac or Linux.

→ Large libraries - Comes with rich libraries like pandas, numpy, matplotlib, etc..

→ object oriented - Supports classes, inheritance, and reusability.

→ Open Source - Free to use & modify.

→ Community Support - Huge global community for learning and troubleshooting.

3) What is the difference b/w python 2 and Python 3?

Feature	Python 2	Python 3
Print statement	<code>print "Hello"</code>	<code>Print("Hello")</code>
Division	$5/2 = 2$	$5/2 = 2.5$
unicode support	Limited	Full unicode support (better for text & emojis)
updates	no longer supported	Actively updated.
Syntax	older & less clean	modern & more readable.

In short :- Python 3 is the latest version - faster, more efficient, and widely used today.

4) What are Python's applications in real-world projects?

Python is used in almost every field today.

- 1) Data science & Analytics - Data cleaning, visualization and prediction.
- 2) Web development - Using frameworks like Django, flask
- 3) Machine learning & AI - Using Tensor-flow, Scikit-learn, PyTorch
- 4) Automation / scripting - Repeating tasks automatically.
- 5) Game Development - Using libraries like pygame.
- 6) IOT (Internet of things) - Connecting smart devices
- 7) Cyber security, chatbots, & cloud computing.

5) What is PEP 8 & why is it important in python Programming?

PEP 8 stands for Python Enhancement Proposal 8.

It gives coding style guidelines for writing clean, readable and consistent Python code.

Importance -

- Helps keep code neat and professional.
- Makes it easier for teams to understand each other's code.
- Improves readability & reduce errors.

6) Who developed Python & in which year was it released?

Python was developed by Guido van Rossum in 1991. He created it as a simple & powerful programming language that focuses on readability & productivity.

7) What do you mean by "dynamically typed" in Python?

Python is dynamically-typed, which means you don't need to declare the data type of a variable before using it.

The interpreter automatically detects the type at runtime.

Ex - `x = 10` # integer

`x = "hello"` # now string, no error!

The type of `x` changes automatically.

8) What is the difference b/w a compiler & an interpreter, & which does Python use?

Feature
Execution

Compiler

Translates the whole code at once.

Interpreter

Translates code line by line.

Speed

Faster execution

Slower execution

Error Detection

Shows all errors after compilation

Stops immediately when an error occurs

ex languages

C, C++

Python, JavaScript